

STATIONARY TCAS INSTALLATION PROCEDURE

Stationary Kavach unit shall comprise of

- a) Station/LC/IB Kavach Vital Computer
- b) Stationary Kavach Radio Unit with 40 Meters tower
- c) Station master operation and indication panel
- d) GPS& GSM Unit
- e) Mono ploe Antennas
- f) TCAS Repeater Relay Rack
- g) Remote interface unit
- h) RFID TAG
- i) Power Distribution Unit
- j) DC DC Converter
- k) Station Termination panel & SM OCIP Termination panel



Station/LC/IB Kavach Vital Computer



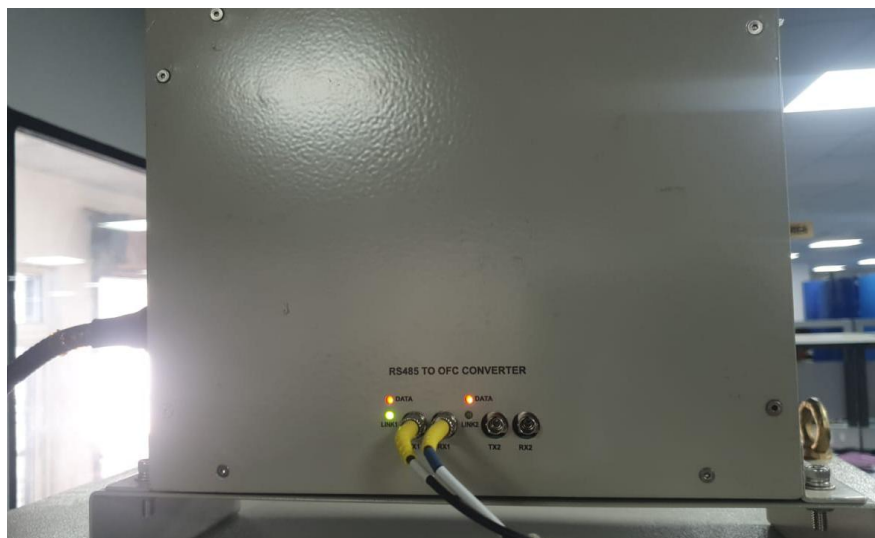
Stationary Kavach Radio Unit



Remote interface unit



Station master operation and indication panel(Front View)



Station master operation and indication panel(Rear View)



Power Distribution Unit(Front View)



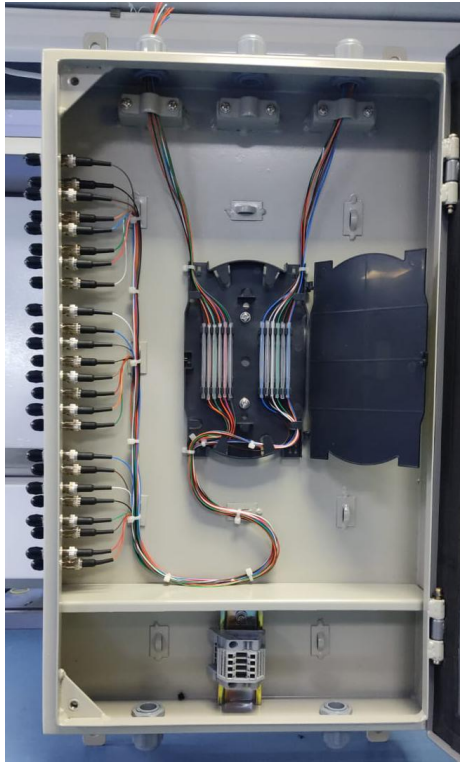
Power Distribution Unit(Inside View)



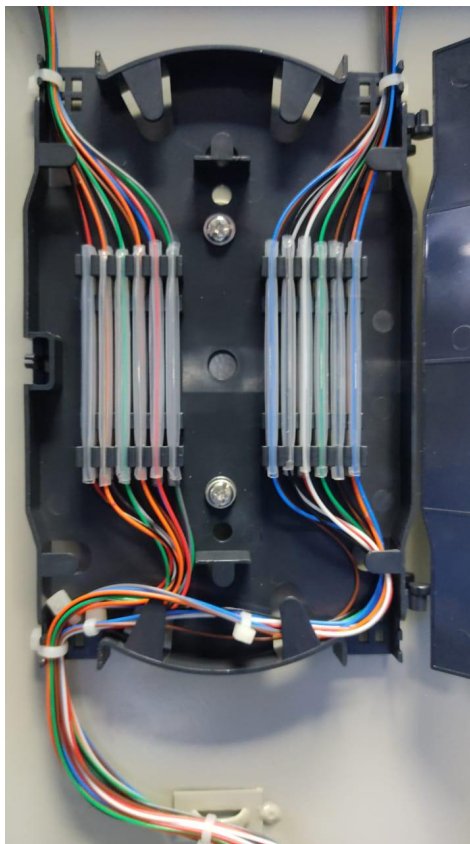
Power Distribution Unit(Installation at site)



Station Termination panel(Installation at site-In Relay room)



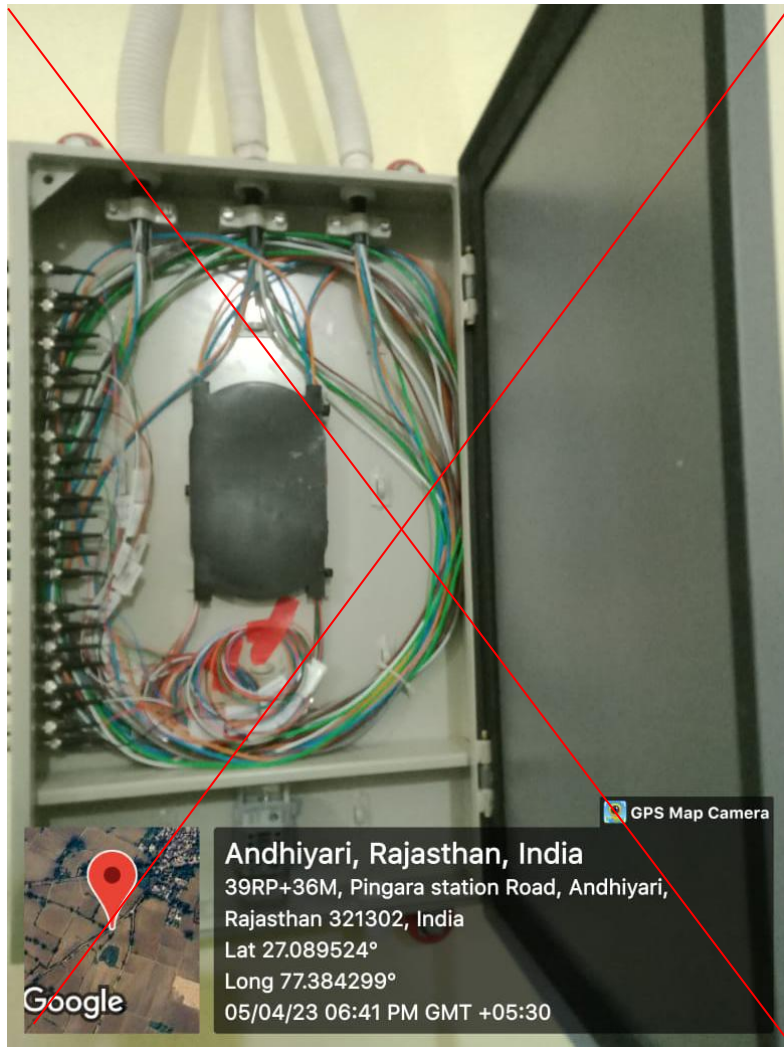
Station OFC Termination panel(Installation at site-In Inside)



Station OFC Termination panel(Installation at site-In Inside)

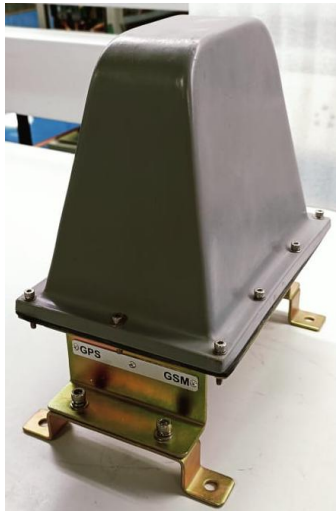


Station OFC Termination panel(Installation at site-In Inside)



Station OFC Termination panel(Installation at site-In Inside)

Not preferably as above.



GPS & GSM Unit



GPS & GSM Installation at site on the top of the roof(I.e Always open to sky)



TCAS REPAEATER RELAY RACK(Installation at site in relay room)



Installation of units in Relay room with Insulators with Anchor bolts



Radio Units with Mono pole antennas in 40 meters tower



GPS Map Camera



Secunderabad, Telangana, India

#135, Archana Arcade ,adj Srikara hospital
Oppo. Railway Reservation Complex, Railway
Colony, Chilakalguda, Secunderabad,
Telangana 500025, India

Lat 17.434886°

Long 78.506428°

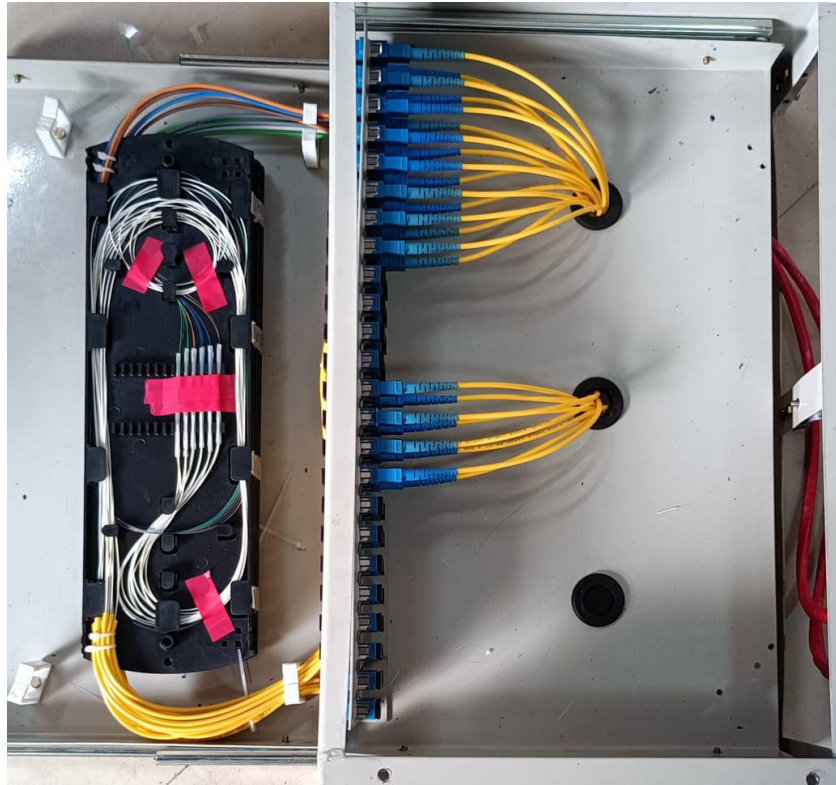
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42" Inch OFC Termination Racks(In Relay room)



17" Inch OFC Termination Racks(In Relay goomty rooms)



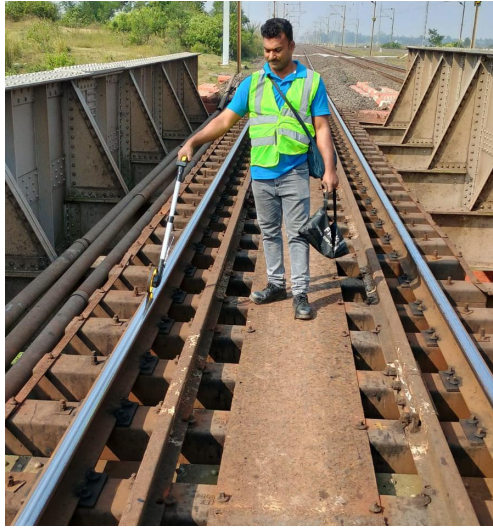
OFC Splicing in inside the OFC Termination unit(FDMS)



DC DC CONVERTERS FOR KAVACH & RELAY RACKS POWER SUPPLY



RIU Installations rooms DC DC Rack and Repeater Relay rack installations



RFID TAG fixing as Layout plan at site



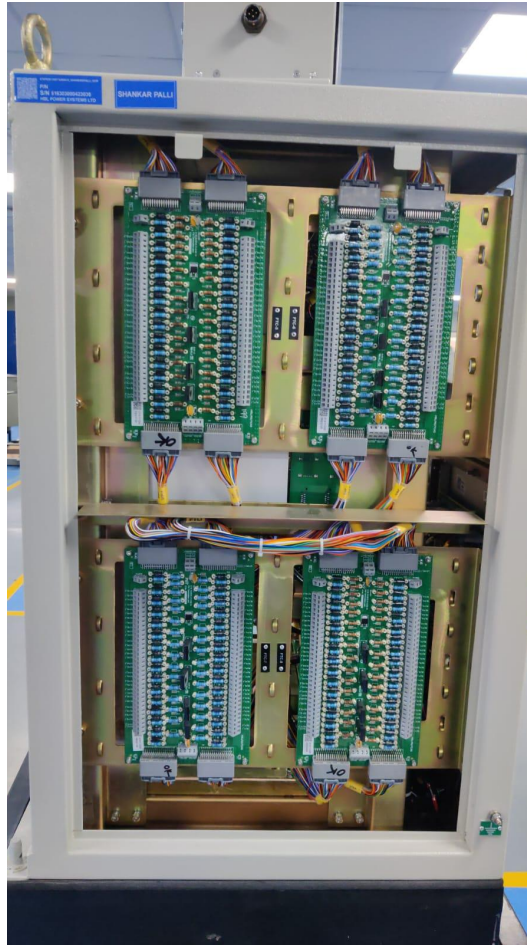
RFID TAG fixing as Layout plan at site



RFID TAG FIXING ON SLEEPERS



RFID TAG FIXING ON SLEEPERS



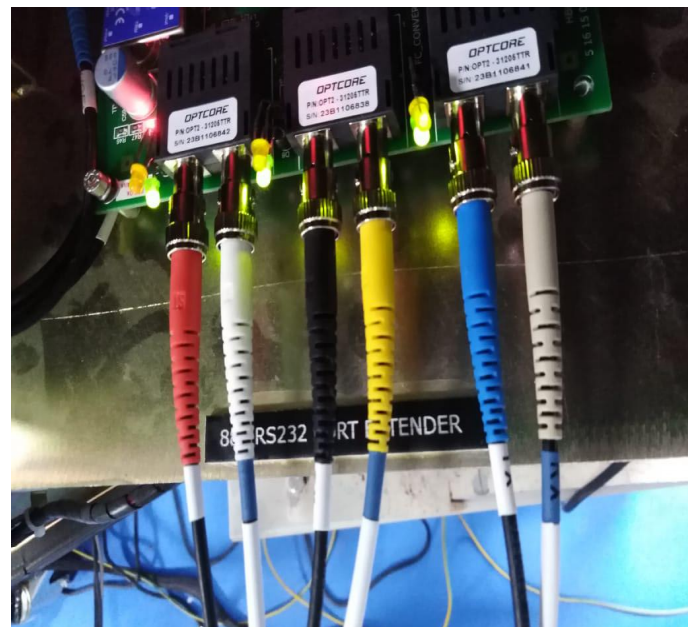
Field unit termination in Stationary kavach in Relay room



Field unit termination in Stationary kavach in Relay room



OFC Terminations in Stationary TCAS side in inside at RS232 -OFC Converter



OFC Terminations in RTU Tower side in inside at RS232 -OFC Converter



RTU Tower Unit



Eathernet to Optical converter Model:5022 Modem(Front View)



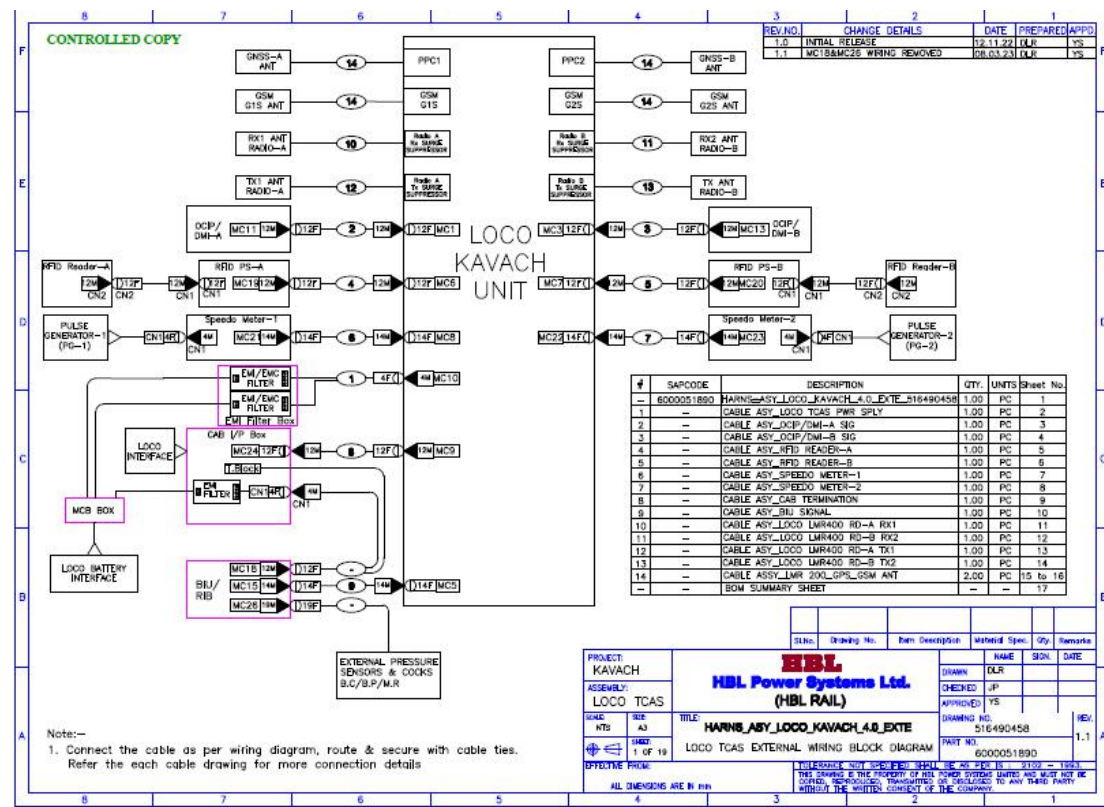
Figure2:Eathernet to Optical converter Model:5022 Modem(Back View)



Complete Stationary Kavach installation at site(Either in relay room /Separate room)

On Board Kavach Installation components

1. On board Kavach
2. Radio Unit
3. GPS GSM Unit
4. Radio Antennas
5. RFID Readers with PSU Interface unit
6. Pulse generators with Speedometer with interface unit
7. Cab termination unit
8. LP OCIP Units
9. RIB/BIU (Brake Interface Unit)
10. MCB EMI Filter unit
11. MCB BOX



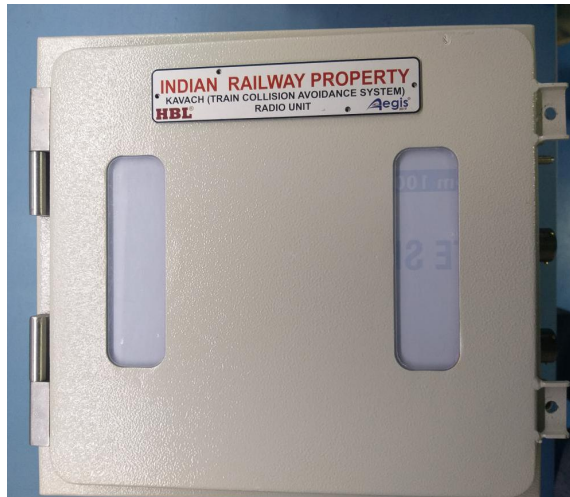
On Board Kavach block diagram



On Board Kavach Unit(Front View)



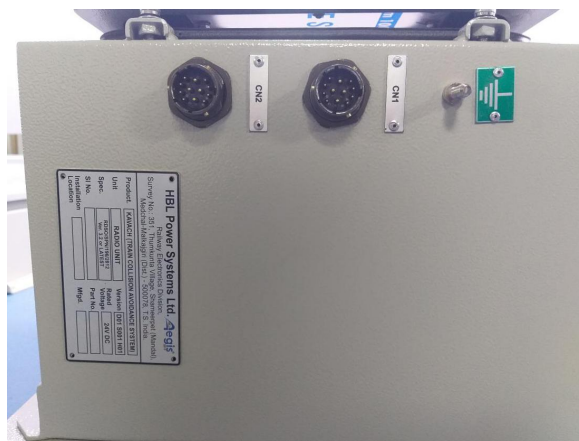
On Board Kavach Unit(Bottom View)



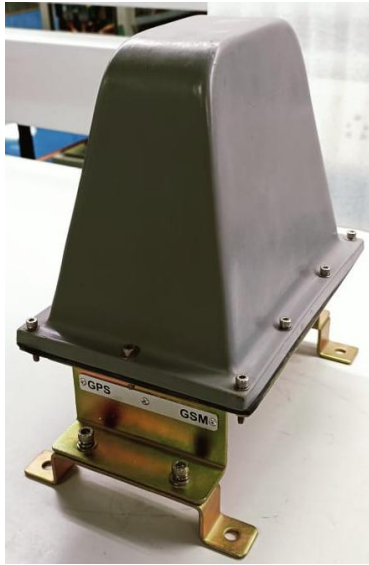
Radio unit(Front View)



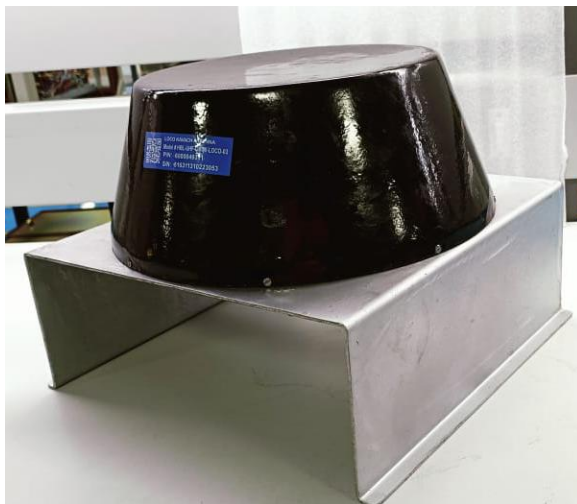
Radio unit(Back View)



Radio unit(Side View)



GPS & GSM Unit



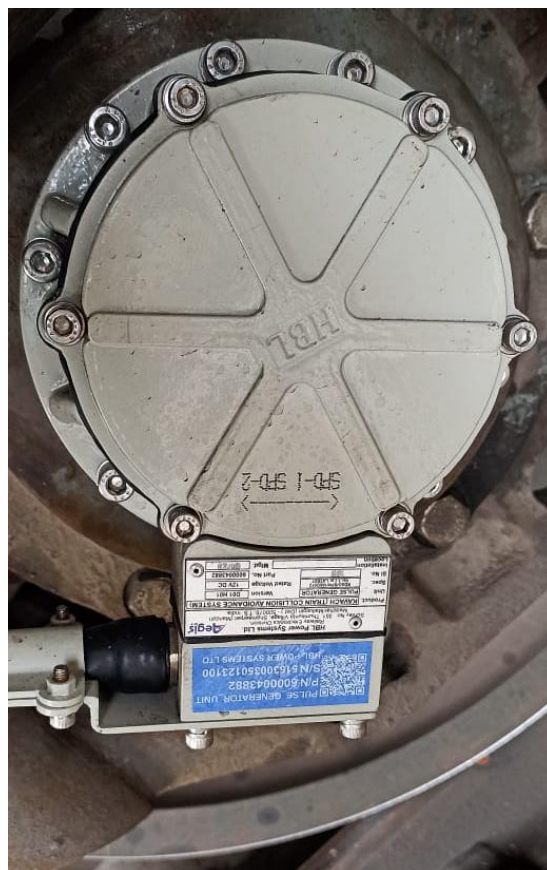
Radio Antennas



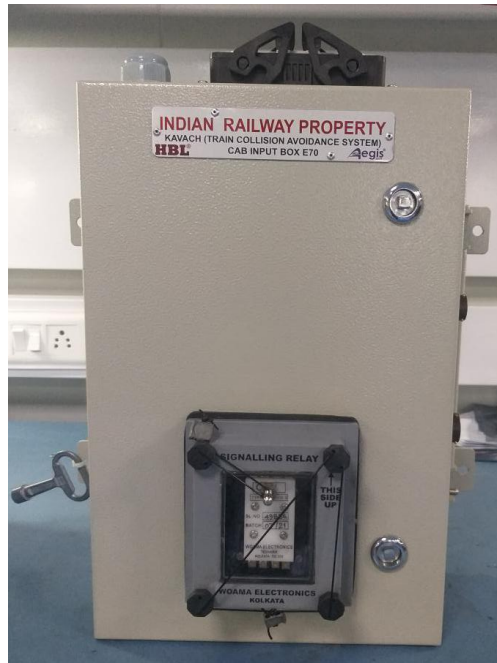
RFID Reader with Interface unit



Speedometer/PG interface units



Pulse Generator



Cab Termination Unit(Front View)



Cab Termination Unit(Side View)



Cab Termination Unit(Inside View)



LP OCIP (Front View)



LP OCIP (Bottom View)



